

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456410.67549 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.019
Transit depth (Rp/Rs)^2	2.68 +/- 0.61 [%]
Orbital Inclination (inc)	86.93 +/- 1.9
Airmass coefficient 1 (a1)	0.983 +/- 0.014
Airmass coefficient 2 (a2)	0.013 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.43 %
Best Comparison Star	#3 - [257, 291]
Optimal Aperture	3.5
Optimal Annulus	14.98
Transit Duration (day)	0.06 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.164 ± 0.019 vs 0.1594 (deviation 0.15σ)
Duration fit vs published	0.06 d vs 0.0512 d (deviation 2.37σ)
Mid-time O-C	1.2 min
Depth SNR	5.4
Residual scatter	1.43 %

Light curve

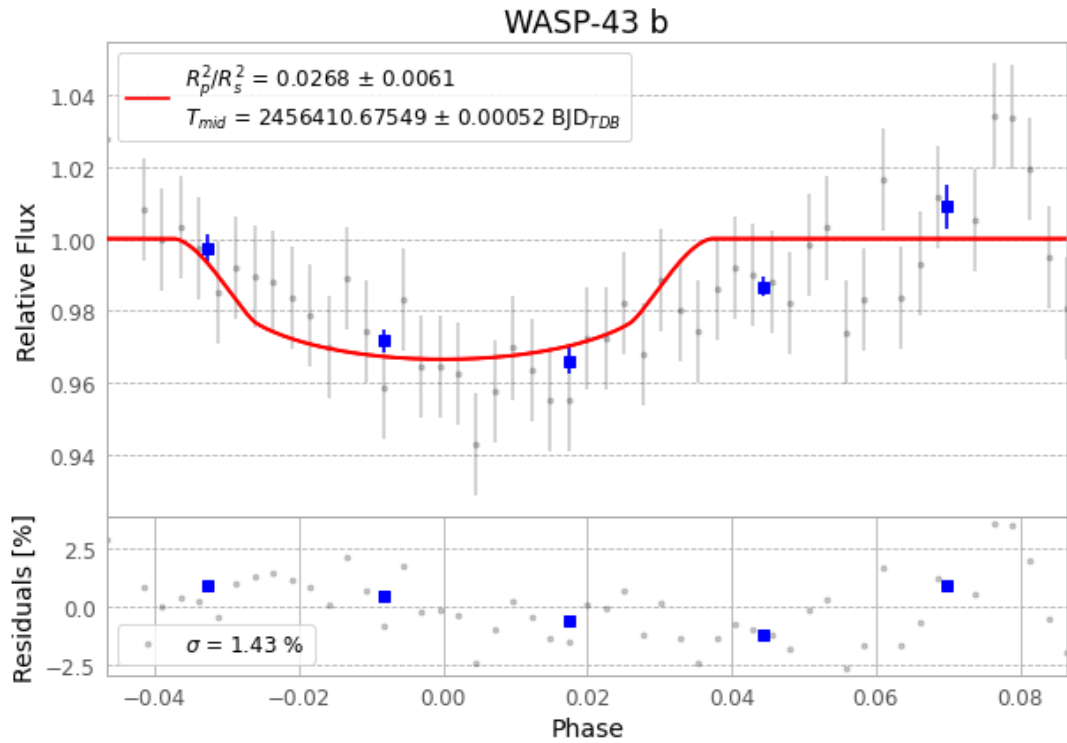


Figure 1 — FinalLightCurve_WASP-43 b_2013-04-27.png (SHA-256 bd8e6bfc970d2f2e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [323, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c0793d425c8e3b4b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), WASP-43130428031212.FITS → WASP-43130428054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130428.log (eae08497b770...), FinalLightCurve_WASP-43 b_2013-04-27.png (bd8e6bfc970d...), FinalLightCurve_WASP-43 b_2013-04-27.csv (9705cd821dde...)

Compute resources

Wall time 117.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 01:13Z.